COMP4442 Service and Cloud Computing

Tutorial 1

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About Me

- Research Interests:
  - Network Monitoring/Measurement
  - Collaborative Edge Computing
  - Service Computing
- Skills:
  - Kubernetes
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Outline

- Definition of Cloud Computing
  - Cloud Computing is Everywhere
  - Cloud Computing is Never New
  - How Others Define Cloud Computing
  - Evolution of Cloud Computing
- Short Introduction to Amazon Web Services (AWS)
  - Why do we use AWS?
  - Brief Introduction to Popular Services
  - Example Application: Hosting Websites
- Some Learning Tips

Cloud Computing is Everywhere

- Did you use QQ or Google Docs?
- Did you use a virtual machine or container?
- How can you backup your photos on your phone?
- How can you develop and deploy software?

Cloud Computing is Never New

- Back to 1963 in MIT, a project about “computer to be used by two or more people, simultaneously.”
- Late 1990s, Salesforce provide a platform for users to download program/applications
  - Vs. install program/applications via floppy disk

- In 2006, the first popular cloud service: AWS EC2
  - Allow individuals to rent virtual computers to use their own application
- Since 2010, Dropbox / Google Drive become popular
- Almost all enterprises provide cloud platforms and services.

How Others Define Cloud Computing

- *Cloud computing is the delivery of different services through the Internet. These resources include tools and applications like data storage, servers, databases, networking, and software.* [From Investopedia]
- *Cloud computing is the on-demand availability of computer system resources, especially data storage (cloud storage) and computing power, without direct active management by the user.* [From Wikipedia]
- *Cloud computing is the delivery of computing services – including servers, storage, databases, networking, software, analytics and intelligence – over the Internet to offer faster innovation, flexible resources and economies of scale.* [From Microsoft Azure]
- *Cloud computing is the on-demand delivery of IT resources over the Internet with pay-as-you-go pricing.* [From Amazon Web Services]

The Best Cloud Computing Definition

- Quote from Microsoft:
  - Cloud computing is the delivery of computing services—servers, storage, databases, networking, software, analytics, intelligence and more—over the Internet ("the cloud") to offer faster innovation, flexible resources and economies of scale. You typically pay only for cloud services you use, helping lower your operating costs, run your infrastructure more efficiently and scale as your business needs change.

- Why "cloud"? Cloud is used as a metaphor for the Internet
  - A cloud-like shape was used to denote a network on telephony schematics and later to depict the Internet in computer network diagrams.
  - An updated version of Utility Computing: basically, virtual servers available over the Internet.
  - Utility Computing: A pool of virtualized computer resources that IT can access on demand.

Five Essential Characteristics

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- On-demand self-service
  - Cloud computing should incorporate sufficient **agility** and **autonomy**, that requests for more resource are automatically and dynamically provisioned in real time, without human intervention.
- Broad network access
  - Capabilities are **available over the network** and accessed through standard mechanisms that promote use by heterogeneous **thin or thick client platforms** (e.g., mobile phones, tablets, and laptops)

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- Elastic resource pooling
  - The provider's computing resources are pooled to serve multiple consumers using **a multitenant model**, with different physical and virtual resources **dynamically assigned and reassigned** according to consumer demand.
- Rapid elasticity
  - Requests for extra resource are self-managed and automatic in **relation to demand**. From the consumer's perspective, the supply of compute resources is limitless.

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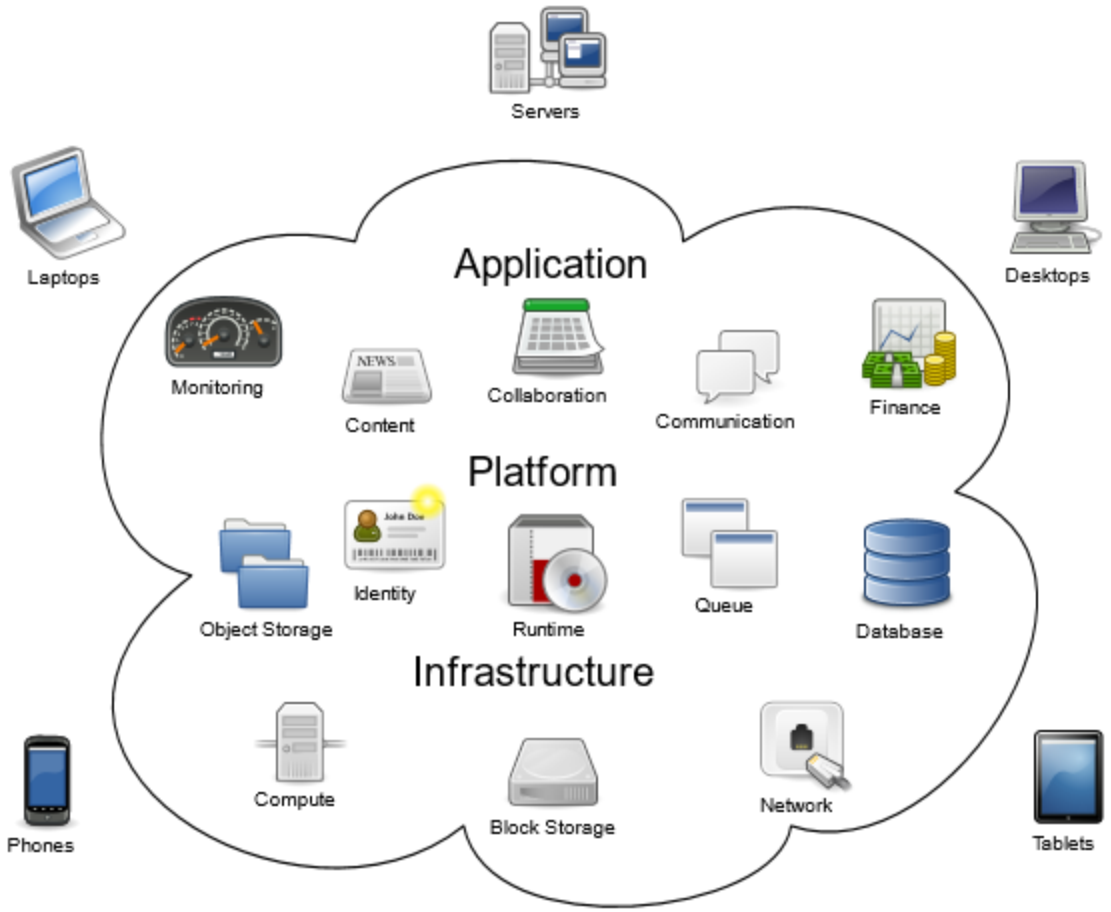
- Measured service
  - In the same way that energy usage can be monitored, controlled and reported, cloud computing resource providers dynamically optimize the underlying infrastructure and provide a transparent metering service at a level of abstraction that is relevant to the consumer.

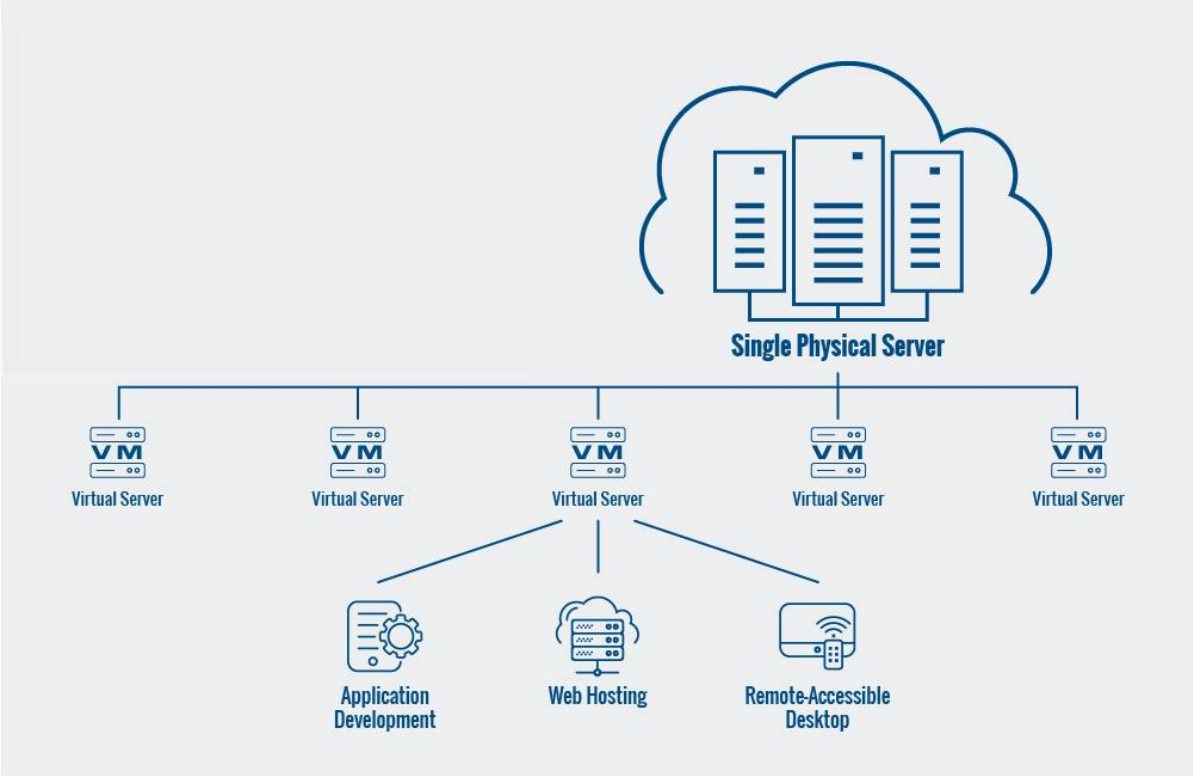
Cloud Computing Services

- Services provided by cloud computing
  - Cloud server
  - Storage, computation(CPU, GPU, FPGA...), networking
  - Development platforms
  - Applications
  - …



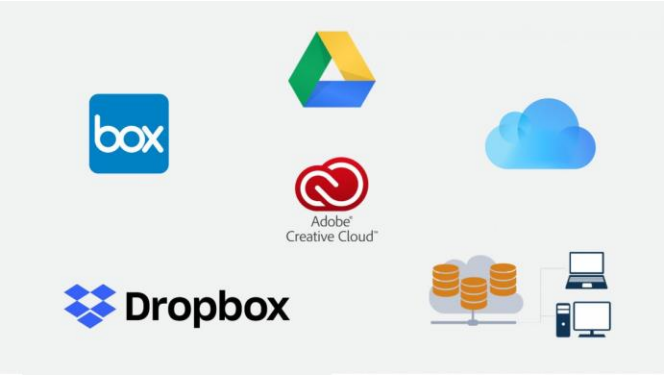
Cloud Computing Resources

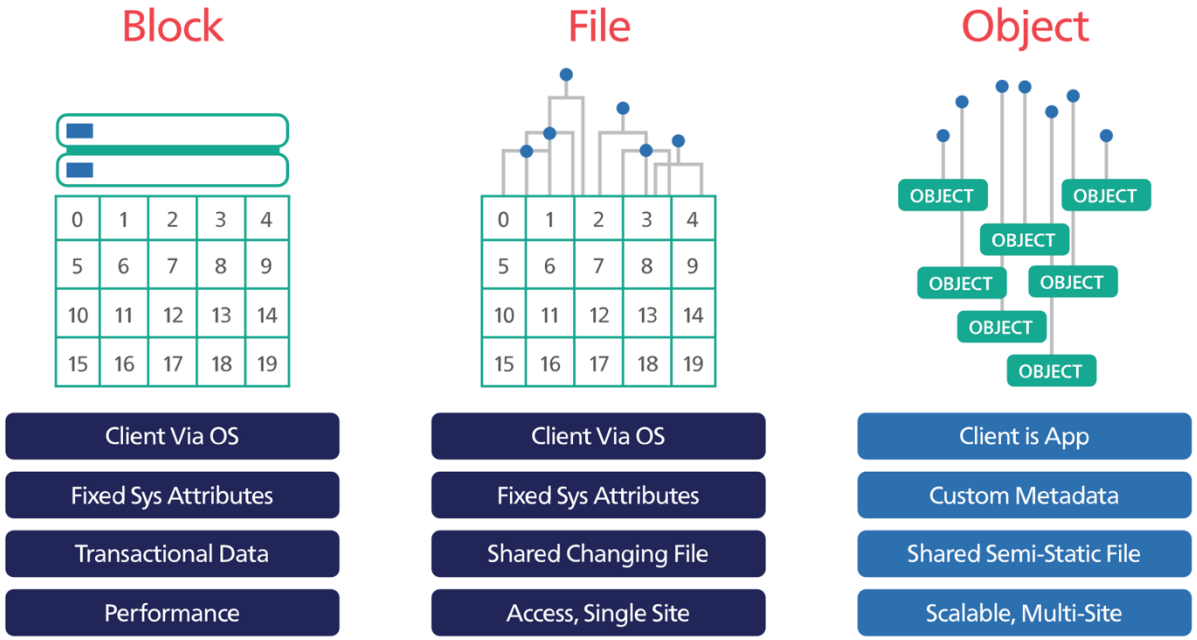
- Cloud server
  - Generally, it is a virtual computer hosted by physical servers
  - Created by virtualization technology over physical machine
  - Running applications, processing data, storage data, etc



Cloud Computing Resources

- Cloud storage
  - Fault tolerance/security
  - Usually integrated with other resources or services in the cloud
  - Object, block, file storage service.
  - Web database (MySQL, levelDB)
  - Cloud disk, cloud backup, cloud synchronize
  - …





Cloud Computing Resources

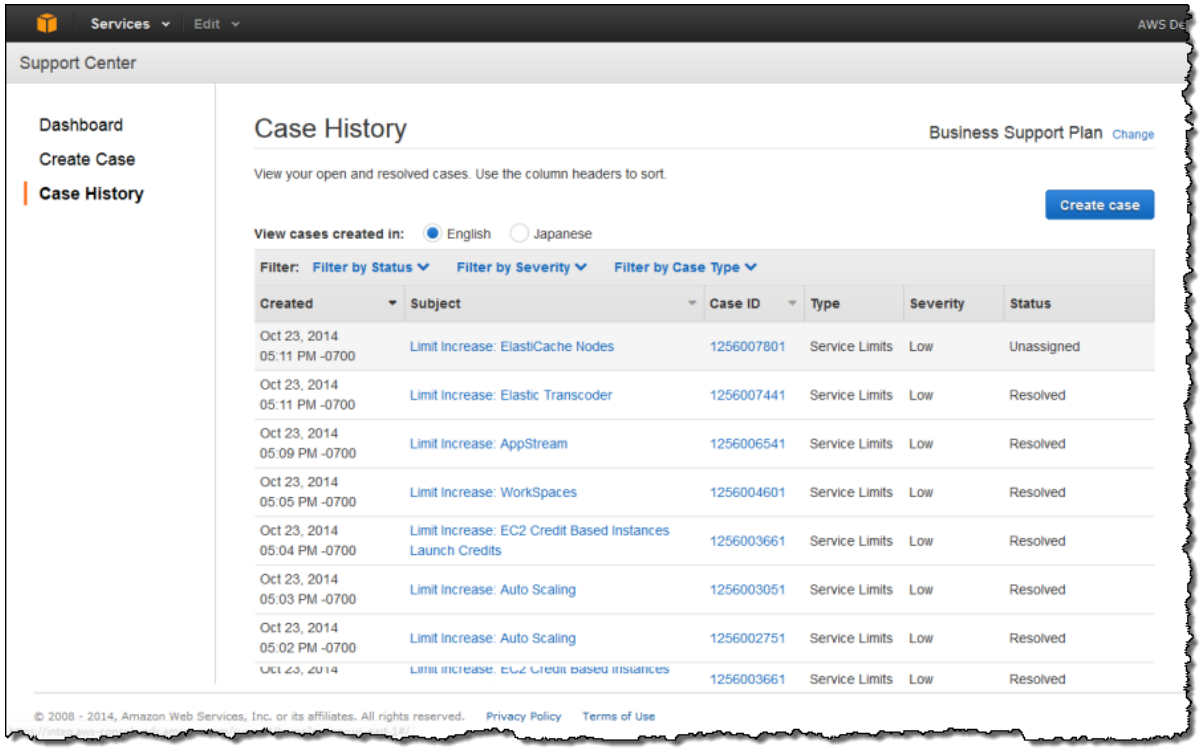
- Cloud computation power
  - Solely use computing resources of the cloud.
  - Use API (application programming interface) to interact with the cloud.
  - Machine learning.
  - Data processing and analyzing.
  - Brute-force searching.
  - …

Cloud Computing Resources

- Cloud application software
  - Ready-made for direct use
  - Web applications
  - Speech/text recognition backend
  - Hosted software
  - Virtual desktop
  - …

Cloud Computing Resources

- Development, deployment, management tools
  - Integrated with other resources and services
  - Programming environments and platforms
  - Batch deployment among multiple servers
  - Cloud console for administration
  - …



Why Cloud Computing?

- Cut cost
  - Companies can minimize or completely cut initial infrastructure/software investment, nor do they need to **hire professionals** for development, operation, and maintenance
- Improve adaptability and flexibly expand it needs
  - Enterprise configure IT equipment according to the peak demand, it will be underutilized ordinary times, resulting in a waste of investment – **over-provisioning**
  - With cloud services, companies can flexibly adjust the resources they purchase at any time
- Data can be accessed instantly anytime, anywhere – Cloud storage
- Security
  - A company maintains its own datacenter needs to keep on top of all developing threats and attacks
  - The maintenance and security are ensured by service providers

Evolution of Cloud Computing

Infrastructure as a Service (IaaS)

- With IaaS, you **rent IT infrastructure** – servers and virtual machines (VMs), storage, networks, operating systems – from a cloud provider on a pay-as-you-go basis

Platform as a Service (PaaS)

- PaaS is a **complete development and deployment environment** in the cloud, with resources that enable customer to deploy customer-created applications

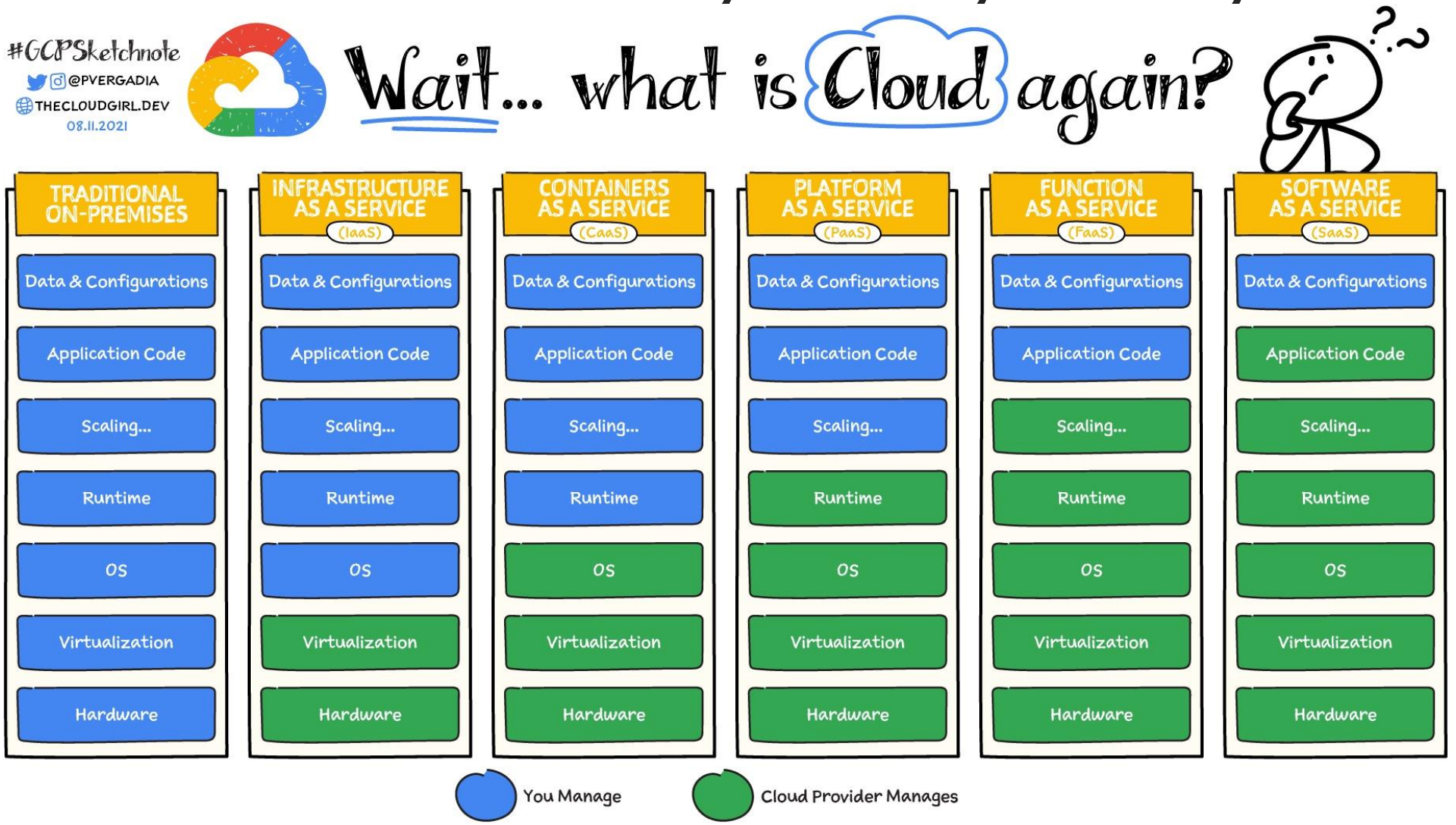
Software as a Service (SaaS)

- **Cloud-based applications** are deployed on a cloud platform and users connect to the application over the Internet. The application is owned by the provider

Evolution of Cloud Computing
-- Other "As A Service" Models

- Many specialized variations of the cloud delivery models exist, each comprised of a distinct combination of IT resources
  - Function as a Service (FaaS)
  - Container as a Service (CaaS)
  - Everything/Anything as a Service (XaaS or *aaS or EaaS)
    - This describes a business model in which a product use is offered as a **subscription-based** service rather than as an artifact owned and maintained by the customer
  - AI as a Service, Backend as a Service, Content as a Service, Desktop as a Service

Difference on IaaS, PaaS, SaaS, CaaS



Infrastructure as a Service

- IaaS allows the cloud user to deploy and run arbitrary applications but does not manage or control the underlying cloud infrastructure
  - The cloud provider owns the equipment and is responsible for housing, cooling operation, and maintenance.
  - The user may have control over the operating systems, storage, and deployed applications and possibly limited control of select networking components (e.g., host firewalls)
- The benefits of IaaS
  - Reduce the need for high, up-front capital expenditures or unnecessary “owned” infrastructure
  - Infrastructure scales and shrink resources to support dynamic workloads, especially in the case of “spiky” workloads
  - Flexible, innovative services available on demand
  - This model gives the most control but also requires the most users to do more set up and management
- Examples: Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform

Infrastructure as a Service

- With IaaS, IT resources are typically virtualized and packaged into bundles that simplify up-front runtime scaling and customization of the infrastructure
- IT resources available through IaaS environments are generally offered as freshly initialized virtual machine instances.
- Virtual machines are leased by specifying hardware requirements, such as processor capacity, memory, and local storage space.

Platform as a Service

PaaS gives the capability to deploy consumer-created or acquired applications using programming languages and tools (application stack) supported by the cloud provider

- The necessary operating system and platform-related tools such as language compilers are **already installed** and maintained by the cloud provider
- Consumers may be able to extend the existing tool set by installing their own tools, but
- absolute control of the infrastructure is still retained by the provider.
- Consumer **has control over application development, deployment and**
- **configuration**, within the confines of the hosted environment.
- This situation has most in common with **traditional web hosting**.
- Examples: AWS Elastic Beanstalk, Amazon API Gateway, Azure App Services, Azure Database for MySQL, and Google App Engine

Platform as a Service

The benefit of PaaS

- Develop and deploy applications and get to market faster - customer only needs to take care of data and application

In general, PaaS solutions have three main parts:

- Cloud infrastructure including virtual machines (VMs), operating system software, storage, networking and firewalls – managed by the cloud provider
- Software for building, deploying and managing applications
- An online GUI where development or DevOps teams can do all their work throughout the entire application lifecycle
- Delivers all standard development tools through the GUI, developers can log in from anywhere to collaborate on projects, test new applications or roll out completed products

Service as a Service

- SaaS is a service model that provides users with subscription-based access to software applications but with no control of the platform or the infrastructure
  - Cloud-based applications run on distant computers “in the cloud” that **connect to** users’ computers via the internet and, usually, **a web browser**.

- The benefits of SaaS
  - You can sign up and rapidly start using the software at or from anywhere
  - without the need of installation
  - No data is lost if your computer breaks, as data is in the cloud
  - The service is able to dynamically scale to usage needs
  - Service providers handle the software updates
  - Usage-based pricing

Serverless Computing

- Serverless ≠ No servers
  - FaaS is the compute model central to serverless and the two terms are often used interchangeably; serverless is more than just FaaS
  - Serverless is an entire stack of services that can respond to specific events or requests and scale to zero when no longer in use
  - Provisioning, management and billing are handled by the cloud provider and invisible to developers
  - Developers can pay less, less attention to infrastructure, and focusing on business logic
- Disadvantages
  - Testing and debugging become more challenging because developers do not have visibility into backend processes
  - Suffer with “Cold Start” performance issue
  - Serverless architectures are not suitable for long-running processes

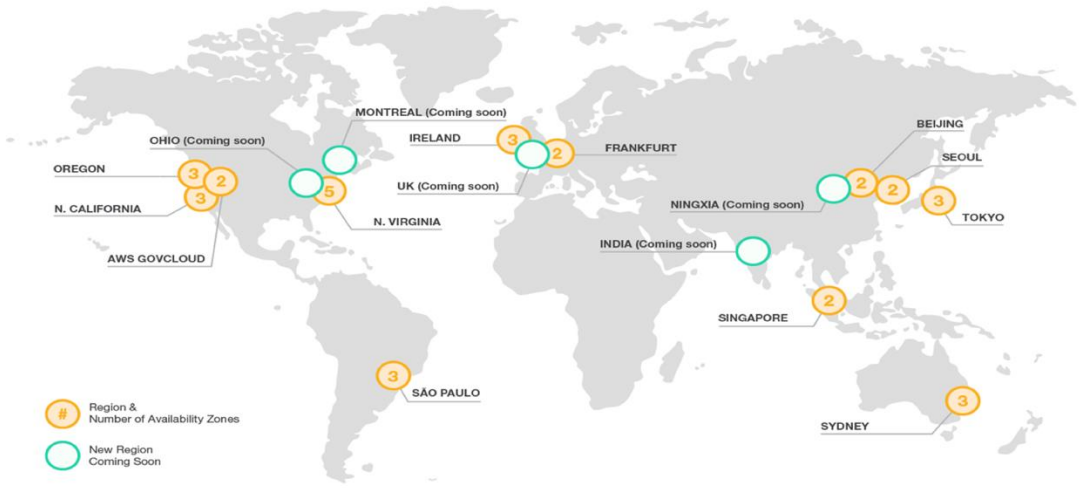
Function as a Service

- FaaS, is a serverless computing model that enables developers to run code in response to specific events or requests
  - Developers can divide the server-side logic into functions that can be scaled automatically and independently
  - Short-lived, provisioning time (charged) in seconds
  - The system dynamically spins up the function in response to a user request
  - When a function is invoked, the cloud provider either executes the function on a running server, or, if there is no server currently running, it spins up a new server to execute the function
  - The user pays only when an action occurs; once the action is done, everything stops, and no costs are incurred

Short Introduction to Amazon Web Services (AWS)

Amazon Web Services (AWS)

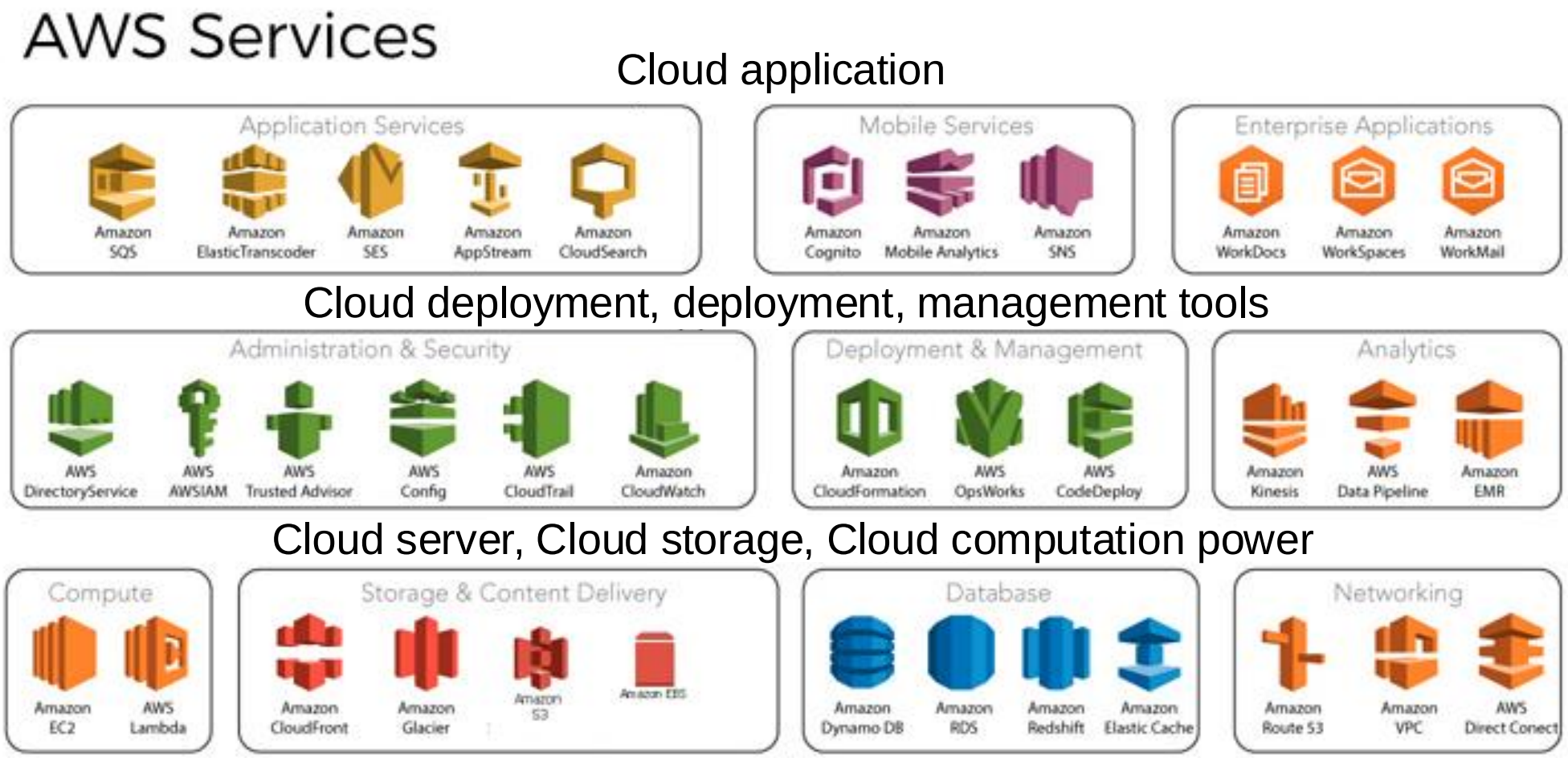
- A collection of various cloud computing services and applications offered by Amazon.com.
- Over 175 fully featured services from data centers globally.



Why AWS?

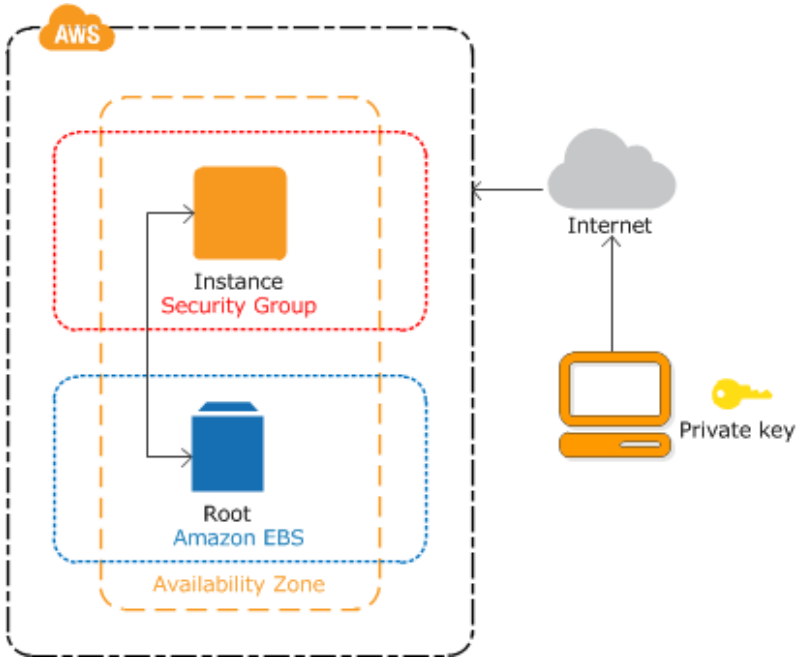
- Something more
  - Data center all around the world -> low latency, high availability, etc.
  - AWS Educate, friendly to students!

AWS Offers


AWS Services
Cloud application
Application Services
Amazon SQS
Amazon ElasticTranscoder
Amazon SES
Amazon AppStream
Amazon CloudSearch
Mobile Services
Amazon Cognito
Amazon Mobile Analytics
Amazon SNS
Enterprise Applications
Amazon WorkDocs
Amazon WorkSpaces
Amazon WorkMail
Cloud deployment, deployment, management tools
Administration & Security
AWS DirectoryService
AWS AWSIAM
AWS Trusted Advisor
AWS Config
AWS CloudTrail
Amazon CloudWatch
Deployment & Management
Amazon CloudFormation
AWS OpsWorks
AWS CodeDeploy
Analytics
Amazon Kinesis
AWS Data Pipeline
Amazon EMR
Cloud server, Cloud storage, Cloud computation power
Compute
Amazon EC2
AWS Lambda
Storage & Content Delivery
Amazon CloudFront
Amazon Glacier
Amazon S3
Amazon EBS
Database
Amazon Dynamo DB
Amazon RDS
Amazon Redshift
Amazon Elastic Cache
Networking
Amazon Route 53
Amazon VPC
AWS Direct Conect

Compute Service: AWS EC2

- Elastic Compute Cloud (Amazon EC2) is a web service that provides secure, resizable compute capacity in the cloud.
- Compute Resources (CPU, RAM, etc.) -> Instance
- A lot of applications demand computation
  - Host a web server dealing with user requests
  - Scientific calculation
  - Building block of other complicated services



Storage Service: S3

- Simple Storage Service (Amazon S3) is an object storage service that offers industry-leading scalability, data availability, security, and performance.
  - 99.999999999% availability (why not one more 9?) of objects across multiple availability zones
  - Objects can be stored in different regions (US West, Asia Pacific) -> low latency, high availability, etc.
- Differences with Dropbox / Google Drive?
  - Dropbox / Google Drive hides the details of storage.
  - S3 can be configured

Database Service: RDS

- How do you host a database if you are developing an application that needs to store structured data
  - On local computer? Only convenient for debug..
  - On a computer in PolyU? Only accessible for PolyU LAN..
  - Host on a public cloud!
- Relational Database Service (RDS) is a database service that makes it easy to set up, operate, and scale a relational database
- Of course, AWS has non-relational database

What Can we Do with AWS

- Platform
  - Amazon elastic compute cloud (EC2)
- Securely store your data
  - Amazon simple storage services (S3)
  - Relational Database Service (RDS)
- Big data management and analytics
  - Amazon elastic MapReduce (EMR)
- Artificial intelligence
  - Amazon recognition
- …

Example: Website Development

- Target: a health information collection website during COVID-19
  - Project of last years' COMP4442

- *this is just part of previous years' project*

Recommendation

- Cloud Computing heavily relies on kubernetes.
  - It is the operating system for cloud computing.

- Cloud Computing has extensive knowledge prerequisite.
  - Operating System
  - Computer Architecture
  - Computing Network
  - Virtualization, Linux, Terminal, Programming Language

Recommendation

- Debugging in cloud platform could be very difficult.
- Try your application locally before moving to cloud!!
- Practice your skill on a local cluster.

```
~ time minikube start
😄 minikube v1.13.0 on Darwin 10.15.6
✨ Using the docker driver based on user configuration
👍 Starting control plane node minikube in cluster minikube
🔥 Creating docker container (CPUs=2, Memory=3892MB) ...
🐳 Preparing Kubernetes v1.19.0 on Docker 19.03.8 ...
🔎 Verifying Kubernetes components...
🌟 Enabled addons: default-storageclass, storage-provisioner
💡 kubectl not found. If you need it, try: 'minikube kubectl -- get pods -A'
🏄 Done! kubectl is now configured to use "minikube" by default

_____________________________________________________________
Executed in   23.96 secs   fish           external
   usr time    1.66 secs  237.00 micros    1.66 secs
   sys time    0.78 secs  943.00 micros    0.78 secs

~
```

Minikube:
run kubernetes locally